

Data Cleaning, DAX & Dashboard Report

Prepared By: Hemarubini

Project Overview

Objective

The objective of this project is to clean, transform, model, and visualize business data using Power BI, Power Query, and DAX. The dashboard provides meaningful insights for business analysis and decision-making.

Dataset Used

- List of Orders.csv
- Order Details.csv
- Sales Target.csv

Data Cleaning & Transformation

The following data preparation steps were completed using Power Query:

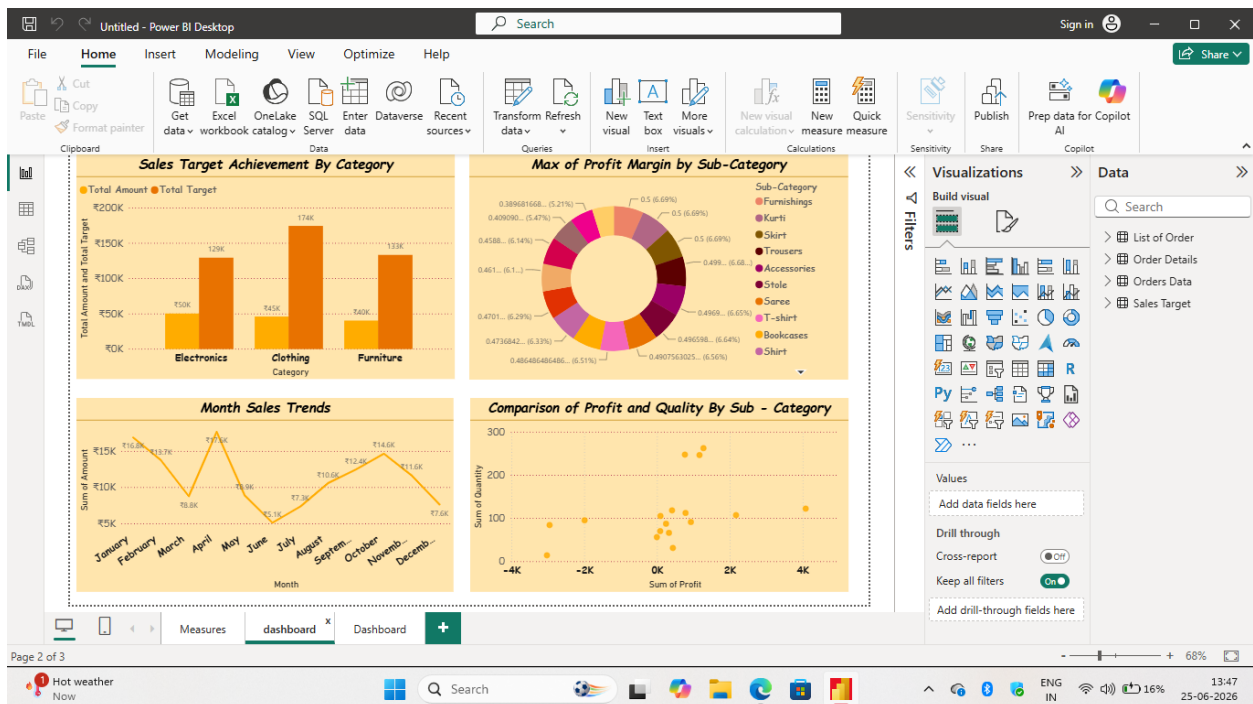
- Imported all datasets
- Changed data types
- Removed duplicates
- Checked missing values
- Created Location column
- Created Profit Margin column
- Created Profit Status column
- Merged tables using Order ID
- Applied sorting and filtering
- Created relationships between tables
- DAX Measures Created

The following measures were created:

- Total Sales
- Total Target
- Total Orders
- Average Profit
- YTD Sales

- Sales Count
- Target Count
- Minimum Target
- Maximum Target
- Average Target

Sales Analysis



Sales Target Achievement by Category

This clustered column chart compares Actual Sales with Sales Targets across different product categories to evaluate target achievement.

Max Profit Margin by Sub-Category

The donut chart displays the maximum profit margin for each sub-category, helping identify the most profitable products.

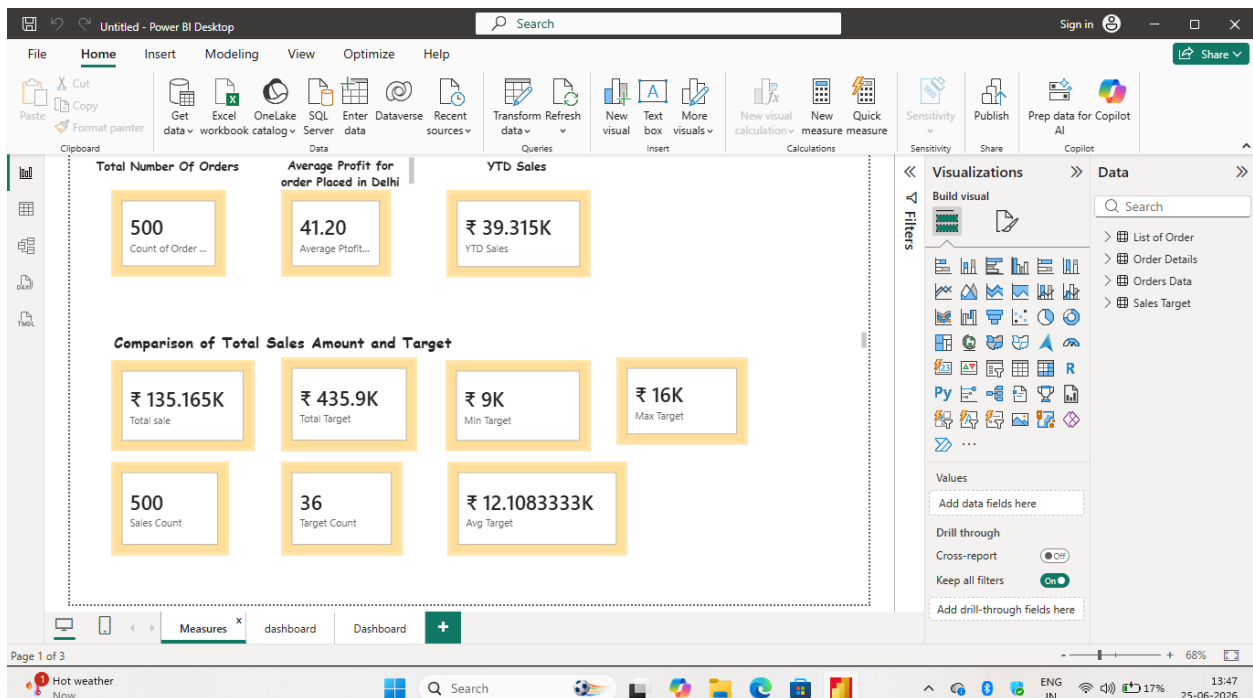
Monthly Sales Trend

The line chart illustrates the monthly sales trend, making it easy to analyze business performance over time.

Profit vs Quantity

The scatter chart compares Profit and Quantity Sold to understand the relationship between sales volume and profitability.

KPI Dashboard



Total Number of Orders

Displays the total number of customer orders.

Average Profit

Shows the average profit earned from orders placed in Delhi.

YTD Sales

Displays cumulative sales from the beginning of the year.

Sales & Target KPI Cards

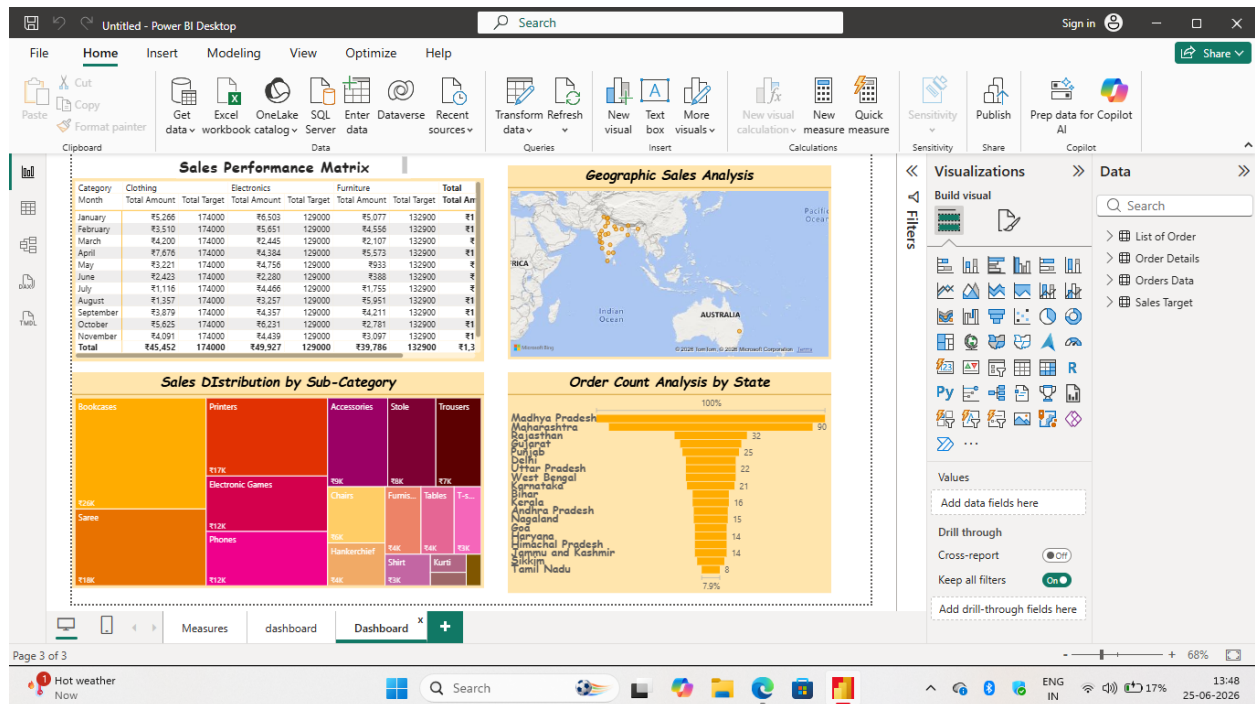
The KPI cards display:

- Total Sales
- Total Target

- Minimum Target
- Maximum Target
- Sales Count
- Target Count
- Average Target

These cards provide a quick summary of business performance.

Business Insights



Sales Performance Matrix

Compares Actual Sales and Sales Targets across different categories and months.

Geographic Sales Analysis

The map visual displays total sales by city to identify regional sales performance.

Sales Distribution by Sub-Category

The treemap represents the sales contribution of each product sub-category.

Order Count Analysis by State

The funnel chart displays the distribution of order counts across different states.

Key Insights

- Sales performance varies across different product categories.
- Some sub-categories generate higher profit margins than others.
- Monthly sales show fluctuations throughout the year.
- Geographic analysis highlights high-performing sales regions.
- KPI cards provide a quick overview of business performance.

Conclusion

This project successfully demonstrates the complete Power BI workflow, including Data Cleaning, Power Query Transformation, Data Modeling, DAX Measure Creation, and Interactive Dashboard Development. The dashboard provides meaningful business insights and supports effective decision-making.